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Background

Posters are important!



Millions of posters are presented annually
One of the most common forms of scientific communication



Typically the first communication of a project
Early stage findings, presented for the first time outside the team



Can increase the pace of discoveries
If shared and made reusable

Problem

The vast majority of knowledge from **posters evaporates when conferences end**.

Studies estimate that only a minority of poster findings are ever published as manuscripts, with reported rates ranging between 1% to 50%. Moreover, it takes on average **2 years** from poster presentation to manuscript publication.

When we looked at poster sharing, we only found **150,000 posters** across 86 platforms although we were expecting to find millions.

Even when posters are shared, **metadata is not provided properly**, making them difficult to find and reuse (**Fig. 1**).

Challenges

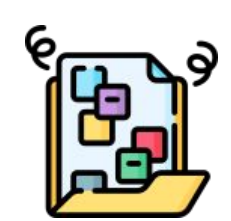
When we asked researchers why they are not sharing or reusing posters consistently, we identified several challenges.



Lack of awareness (why/how to share posters?)



Time consuming and no incentives



Unstructured (large scale analyses are difficult)

Solution: Posters.science

To address these challenges, we are developing Posters.science, an open-source (MIT license) and free platform for **easily sharing and finding FAIR and AI-ready posters**.

The platform integrates LLM-based tools to auto-extract metadata from posters, making it easier for researchers to **share metadata-rich posters**.

Posters.science also operates as a registry by indexing posters shared on different platforms to provide a central location for **discovering them conveniently**.

Metadata	Zenodo	Figshare
Title	100%*	100%*
At least one keyword	54%	100%*
Funding information	20%	28%
ORCID for all authors	26%	14%
At least one reference	8%	0%
Conference name	47%	NA
Conference dates	45%	NA

* = mandatory

Figure 1. Percentage of posters from Zenodo and Figshare containing a given structured metadata. The analysis is based on 24,738 and 14,387 resources marked as posters shared on Zenodo and Figshare, respectively, as of December 2024.

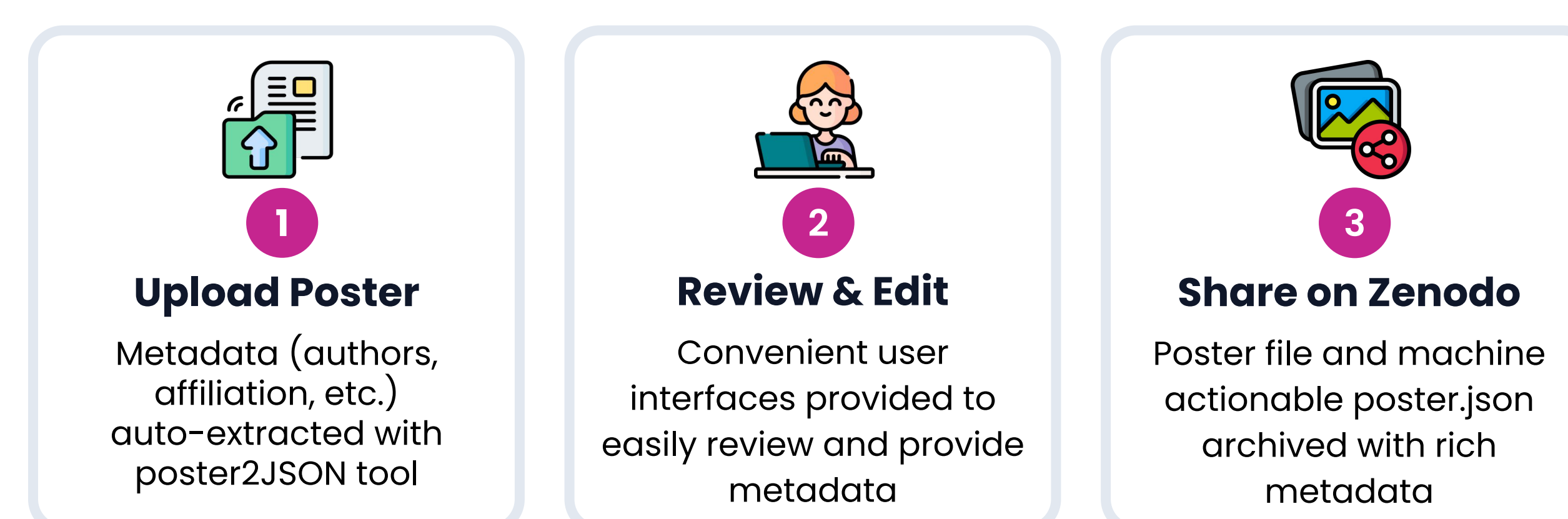


Figure 2. The 3 steps process for sharing posters on Zenodo implemented in Posters.science.

```
{
  "titles": [
    {
      "title": "Toward the Envision Portal: Designing a FAIR and AI-Ready ..."
    }
  ],
  "descriptions": [
    {
      "description": "Ophthalmic imaging datasets are a valuable resource ..."
      "descriptionType": "Other"
    },
    {
      "description": "Purpose\n\nOphthalmic imaging datasets generated in ..."
      "descriptionType": "Abstract"
    }
  ],
  "dates": [
    {
      "date": "2026-05-27",
      "dateType": "Submitted",
      "dateInformation": "Submitted to Zenodo through Posters.science"
    },
    {
      "date": "2026-05-06",
      "dateType": "Presented"
    }
  ],
  "types": {
    "resourceType": "Conference Poster",
    "resourceTypeGeneral": "Poster"
  }
}
```

Figure 3. Example of a poster.json file generated by poster2JSON following our DataCite-inspired poster JSON schema to ensure posters are AI-ready and interoperable.

Results

The platform is now live at **posters.science**. It is built with Nuxt 4, Vue 3, and Nitro 2. PostgreSQL 18 is used as our database.

We have implemented a 3-step process for sharing FAIR and AI-ready posters on Zenodo (**Fig. 2**). We have developed **poster2JSON**, an LLM based Python tool to extract structured metadata from posters and make sharing metadata-rich poster easy.

The metadata is extracted following a **poster JSON schema** we have developed based on the DataCite schema (**Fig. 3**). A poster.json file following that schema is included with each shared poster to make them AI-ready and interoperable.

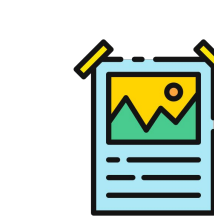
We have indexed over **24,000 posters** from Zenodo and Figshare. Since many records labeled as posters were not necessarily posters, we developed **PosterSentry**, a tool to detect true posters from abstract, conference papers, etc. This allowed us to exclude around 15,000 records from Zenodo and Figshare that were likely not posters.

Conclusion

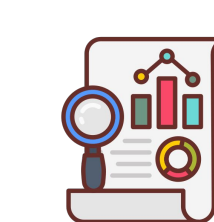
With Posters.science, we are aiming to **address all the challenges of poster sharing and reuse** by providing a single place to



Learn about why and how to share posters



Share FAIR and AI-ready posters easily



Discover interoperable posters conveniently



Share your posters today using Posters.science!

Future work

We are looking to improve posters.science based on **feedback from our users**.

Additionally, we are in the process of **indexing more posters** from different platforms to provide a larger set of posters that can be reused at scale for new discoveries.

We are also expanding the performance and use cases of PosterSentry and poster2JSON in **collaboration with COMET**, a large scale effort to improve metadata quality. We expect these tools to eventually be used to enhance poster metadata directly at the source in the DataCite registry.

